

Vulnerability Assessment Report

Task 2 — Network Security & Scanning | ApexPlanet Cybersecurity & Ethical Hacking Internship

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Date	6-Sep-26
Target scope	Metasploitable2 (isolated Host-Only lab network)
Scanner used	OpenVAS / Nessus Essentials
Timeline	Days 13-24

1. Executive Summary

A vulnerability assessment was conducted against a single Metasploitable2 virtual machine, isolated on a Host-Only lab network with no external connectivity, as part of Task 2 of the ApexPlanet Cybersecurity & Ethical Hacking Internship. The scan was performed using **[OpenVAS / Nessus Essentials]** on **[date]**, following the reconnaissance and Nmap scanning documented earlier in this repository.

The assessment identified **[18] total findings**, comprising **[2] Critical, [3] High, [5] Medium, and [8] Low** severity issues. The most significant risks were concentrated in outdated network services exposed by the target — including **[e.g. "an unauthenticated backdoor in vsftpd 2.3.4," "weak default credentials on the MySQL and Tomcat services," "an insecure Java RMI configuration allowing remote code execution"]** — several of which could allow an attacker to gain full remote access with minimal effort.

These results indicate the target's overall security posture is **[poor / weak by design, as expected of an intentionally vulnerable training host]**. The findings below are prioritized by severity, and the highest-risk issues (**[VULN-01, VULN-02, ...]**) should be treated as immediate remediation priorities. The vulnerabilities identified here directly inform the exploitation work planned for Task 4.

2. Scope & Methodology

This assessment was performed against a single Metasploitable2 virtual machine, isolated on a VirtualBox Host-Only Adapter network with no external connectivity, as part of Task 2 of the ApexPlanet internship. Testing followed an unauthenticated vulnerability scan using OpenVAS (or

Nessus Essentials), preceded by the Nmap reconnaissance documented in *nmap_report.txt* and *nmap-analysis.md* in this same repository.

3. Findings Summary

Severity	Count	Description
Critical	[2]	Immediate risk of full compromise
High	[3]	Significant risk, exploitable with moderate effort
Medium	[5]	Real risk, usually needs another weakness chained with it
Low	[8]	Minor exposure or defense-in-depth issue

4. Detailed Findings

One worked example is included below purely so the format is clear. Delete it and add one entry per real finding from your own scan.

ID	VULN-01
Title	vsftpd 2.3.4 Backdoor Command Execution
Severity	Critical
CVE	CVE-2011-2523
Affected host:port	192.168.56.102:21
Description	The bundled vsftpd 2.3.4 on Metasploitable2 contains a known backdoor that can open a listening command shell. It is intentionally included in this image for training purposes.
Recommendation	Upgrade to a patched vsftpd release. This instance exists only for lab practice and must never be exposed outside the isolated network.

5. Remediation Priorities

How to prioritize:

Severity first — every Critical finding outranks every High, every High outranks every Medium, and so on. Start there.

Within the same severity, ease-of-exploitation breaks the tie — a known backdoor or default credentials (zero skill needed to abuse) outranks something that needs a chained multi-step attack, even at equal severity.

Fixability matters too — a one-line config change or version upgrade can jump ahead of something requiring an architecture change, since it's a priority you can act on immediately.

6. Conclusion

This assessment confirmed the target is highly vulnerable, with the most serious exposures being and These findings will be carried into Task 4, where will be exploited under controlled conditions to demonstrate real-world impact